



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: GRID-INDIA/NLDC/SO/Daily PSP Report

दिनांक: 14.12.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 13.12.2024.

महोदय/Dear Sir,

आईईजीसी-2023 की धारा स.-38(1) के प्रावधान के अनुसार, दिनांक 13-दिसंबर-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 38(1) of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 13.12.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day

Date of Reporting: 14-Dec-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 19:00 hrs; from RLDCs)	57035	62420	42226	20913	2662	185256
Peak Shortage (MW)	0	0	0	0	0	0
Energy Met (MU)	1232	1477	1045	425	50	4228
Hydro Gen (MU)	114	37	73	16	14	254
Wind Gen (MU)	26	206	56	-	-	287
Solar Gen (MU)*	154.61	99.39	63.59	2.91	1.07	322
Energy Shortage (MU)	0.69	0.00	0.00	0.00	0.00	0.69
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	66850	74029	52015	21481	2840	213941
Time Of Maximum Demand Met	10:13	09:46	09:31	17:59	17:16	09:40

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.043	0.00	0.22	7.22	7.44	77.79	14.77

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	9190	0	163.3	78.9	-0.2	583	0.00
	Haryana	8923	0	161.3	105.4	-1.2	504	0.00
	Rajasthan	18212	0	329.3	113.5	-5.0	682	0.00
	Delhi	4463	0	76.7	77.8	-2.5	298	0.00
	UP	19246	0	343.9	123.0	-2.7	567	0.00
	Uttarakhand	2442	0	45.4	34.4	0.0	50	0.00
	HP	2173	0	39.6	32.6	-0.6	140	0.00
	J&K(UT) & Ladakh(UT)	3132	0	63.8	57.9	0.2	585	0.69
	Chandigarh	258	0	4.2	4.2	0.0	26	0.00
WR	Railways_NR ISTS	205	0	4.1	3.8	0.3	37	0.00
	Chhattisgarh	4774	0	96.3	36.5	0.4	403	0.00
	Gujarat	21425	0	405.6	146.5	-4.9	830	0.00
	MP	18555	0	335.4	201.6	-4.1	588	0.00
	Maharashtra	26965	0	562.7	194.8	-5.8	596	0.00
	Goa	740	0	14.9	14.2	0.2	92	0.00
	DNHDDPDCL	1287	0	29.9	30.0	-0.1	37	0.00
	AMNSIL	814	0	18.2	8.5	0.4	274	0.00
	BALCO	529	0	12.6	12.7	-0.1	9	0.00
SR	RIL JAMNAGAR	60	0	1.0	1.0	0.0	0	0.00
	Andhra Pradesh	9697	0	194.2	61.6	-1.0	396	0.00
	Telangana	12891	0	242.6	144.9	0.2	1056	0.00
	Karnataka	12949	0	229.2	54.8	-1.3	456	0.00
	Kerala	4087	0	83.8	68.2	1.0	324	0.00
	Tamil Nadu	14504	0	286.2	168.0	-1.3	711	0.00
	Puducherry	418	0	9.0	8.6	0.0	47	0.00
ER	Bihar	4852	0	90.3	77.5	0.6	244	0.00
	DVC	3224	0	68.0	-41.7	-0.2	301	0.00
	Jharkhand	2020	0	38.5	27.4	-0.4	210	0.00
	Odisha	4529	0	86.1	14.0	-2.7	412	0.00
	West Bengal	7117	0	139.9	12.4	-1.6	25	0.00
	Sikkim	116	0	1.9	1.9	0.0	190	0.00
	Railways_ER ISTS	17	0	0.2	0.2	0.0	0	0.00
NER	Arunachal Pradesh	165	0	2.8	2.8	-0.1	41	0.00
	Assam	1537	0	27.4	19.9	0.5	200	0.00
	Manipur	247	0	3.9	3.8	0.1	50	0.00
	Meghalaya	388	0	6.8	6.0	-0.2	27	0.00
	Mizoram	145	0	2.2	1.8	-0.1	23	0.00
	Nagaland	181	0	2.4	2.4	-0.1	37	0.00
	Tripura	232	0	4.2	3.8	0.0	24	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	-5.7	1.1	-12.8	-12.0
Day Peak (MW)	-500.0	-178.3	-550.0	-717.0

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	256.9	-254.9	198.7	-208.4	7.7	0.0
Actual(MU)	246.6	-268.8	207.5	-199.4	6.1	-8.0
O/D/U/D(MU)	-10.3	-13.9	8.8	9.1	-1.6	-8.0

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	4739	10759	8758	3838	518	28612	52
State Sector	8233	11699	5569	1071	248	26819	48
Total	12972	22458	14327	4909	765	55431	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	693	1429	594	677	9	3403	74
Lignite	19	8	23	0	0	50	1
Hydro	114	37	73	16	14	254	5
Nuclear	32	68	65	0	0	164	4
Gas, Naptha & Diesel	2	11	6	0	23	42	1
RES (Wind, Solar, Biomass & Others)	206	309	146	6	1	668	15
Total	1066	1863	906	699	47	4581	100

Share of RES in total generation (%)	19.35	16.60	16.07	0.88	2.28	14.60
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	33.04	22.24	31.27	2.50	31.05	23.64

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.015
Based on State Max Demands	1.040

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	213941	9:40	62
Non-Solar hr	197035	18:21	0

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 06:00 to 18:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 14-Dec-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	0	0.0	0.0	0.0
2	HVDC	PUSAULI B/B	-	0	251	0.0	6.0	-6.0
3	765 kV	GAYA-VARANASI	2	0	960	0.0	13.4	-13.4
4	765 kV	SASARAM-FATEHPUR	1	28	489	0.0	7.0	-7.0
5	765 kV	GAYA-BALIA	1	0	696	0.0	12.7	-12.7
6	400 kV	PUSAULI-VARANASI	1	0	196	0.0	3.7	-3.7
7	400 kV	PUSAULI -ALLAHABAD	1	0	122	0.0	2.3	-2.3
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	0	597	0.0	9.0	-9.0
9	400 kV	PATNA-BALIA	2	0	1038	0.0	19.1	-19.1
10	400 kV	NAUBATPUR-BALIA	2	0	511	0.0	6.2	-6.2
11	400 kV	BIHARSHARIFF-BALIA	2	9	256	0.0	5.4	-5.4
12	400 kV	MOTIHARI-GORAKHPUR	2	0	543	0.0	9.6	-9.6
13	400 kV	BIHARSHARIFF-VARANASI	2	0	379	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	0	67	0.0	0.5	-0.5
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.1	0.0	0.1
16	132 kV	GARWAH-RIHAND	1	30	0	0.5	0.0	0.5
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						0.7	94.9	-94.3
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	325	1322	0.0	12.4	-12.4
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	422	1314	0.0	6.2	-6.2
3	765 kV	JHARSUGUDA-DURG	2	0	890	0.0	14.2	-14.2
4	400 kV	JHARSUGUDA-RAIGARH	4	0	756	0.0	11.4	-11.4
5	400 kV	RANCHI-SIPAT	2	0	0	0.0	0.0	0.0
6	220 kV	BUDHIPADAR-RAIGARH	1	0	139	0.0	1.7	-1.7
7	220 kV	BUDHIPADAR-KORBA	2	68	101	0.0	0.2	-0.2
ER-WR						0.0	46.2	-46.2
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	316	0.0	6.2	-6.2
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	1992	0.0	46.9	-46.9
3	765 kV	ANGUL-SRIKAKULAM	2	0	2922	0.0	55.7	-55.7
4	400 kV	TALCHER-I/C	2	171	971	0.0	7.2	-7.2
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	108.8	-108.8
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	260	14	2.5	0.0	2.5
2	400 kV	ALIPURDUAR-BONGAIGAON	2	457	0	5.6	0.0	5.6
3	220 kV	ALIPURDUAR-SALAKATI	2	83	0	1.1	0.0	1.1
ER-NER						9.2	0.0	9.2
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	667	0	15.8	0.0	15.8
NER-NR						15.8	0.0	15.8
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	3018	0.0	45.4	-45.4
2	HVDC	VINDHYACHAL B/B	-	0	55	0.0	1.2	-1.2
3	HVDC	MUNDRA-MOHINDERGARH	2	0	983	0.0	24.2	-24.2
4	765 kV	GWALIOR-AGRA	2	0	1920	0.0	27.4	-27.4
5	765 kV	GWALIOR-PHAGI	2	614	1416	1.5	16.2	-14.7
6	765 kV	JABALPUR-ORAI	2	0	800	0.0	24.6	-24.6
7	765 kV	GWALIOR-ORAI	1	803	0	10.8	0.0	10.8
8	765 kV	SATNA-ORAI	1	0	1013	0.0	19.7	-19.7
9	765 kV	BANASKANTHA-CHITORGARH	2	933	1426	2.6	14.3	-11.8
10	765 kV	VINDHYACHAL-VARANASI	2	0	2186	0.0	38.7	-38.7
11	400 kV	ZERDA-KANKROLI	1	189	200	0.0	0.7	-0.7
12	400 kV	ZERDA -BHINMAL	1	131	138	0.5	1.0	-0.5
13	400 kV	VINDHYACHAL -RIHAND	1	976	0	20.8	0.0	20.8
14	400 kV	RAPP-SHUJALPUR	2	587	244	3.9	1.0	2.9
15	400 kV	NEEMUCH-Chittorgarh	2	647	241	4.5	0.6	3.9
16	220 kV	BHANPURA-RANPUR	1	0	204	0.0	3.4	-3.4
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	1.7	-1.7
18	220 kV	MEHGAON-AURAIYA	1	113	0	1.1	0.0	1.1
19	220 kV	MALANPUR-AURAIYA	1	77	12	0.5	0.0	0.5
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						46.2	220.0	-173.8
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	1006	0.0	24.0	-24.0
2	HVDC	RAIGARH-PUGALUR	2	0	4007	0.0	60.9	-60.9
3	765 kV	SOLAPUR-RAICHUR	2	666	1309	1.7	8.4	-6.7
4	765 kV	WARDHA-NIZAMABAD	2	0	2334	0.0	33.3	-33.3
5	765 kV	WARORA-WARANGAL(NEW)	2	0	2561	0.0	40.1	-40.1
6	400 kV	KOLHAPUR-KUDGI	2	1329	0	21.9	0.0	21.9
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	0	0.0	0.0	0.0
9	220 kV	XELDEM-AMBEWADI	1	0	123	2.2	0.0	2.2
WR-SR						25.8	166.7	-140.8

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)	
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	153	-73	-5	-0.12
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	215	-197	-18	-0.43
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	-237	-135	-191	-4.58
	NER	132kV GELEPHU-SALAKATI	-24	-1	-15	-0.36
	NER	132kV MOTANGA-RANGIA	-32	0	-8	-0.19
NEPAL	NR	NEPAL IMPORT (FROM UP)	-66	0	0	0.00
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	-66	0	-36	-0.87
	ER	NEPAL IMPORT (FROM BIHAR)	0	0	0	0.00
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	-180	0	83	2.00
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-498	-486	-493	-11.82
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-717	-400	-501	-12.01
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-52	0	-40	-0.96

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	6.35	0.00	0.00	0.14	0.00	0.00	6.49
Nepal	0.35	0.00	1.23	0.00	0.00	1.00	0.00	0.00	2.58
Bangladesh	12.24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12.24
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	12.59	0.00	7.58	0.00	0.00	1.14	0.00	0.00	21.31

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.68
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.68

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.68	0.00	-6.35	0.00	0.00	-0.14	0.00	0.00	-5.81
Nepal	-0.35	0.00	-1.23	0.00	0.00	-1.00	0.00	0.00	-2.58
Bangladesh	-12.24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-12.24
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	-11.91	0.00	-7.58	0.00	0.00	-1.14	0.00	0.00	-20.63

15 Min (INSTANTANEOUS) ALL INDIA GRID FREQUENCY, GENERATION & DEMAND MET (SCADA DATA)												
TIME	FREQUENCY (Hz)	DEMAND MET (MW)	NUCLEAR (MW)	WIND (MW)	SOLAR (MW)	HYDRO** (MW)	GAS (MW)	THERMAL (MW)	OTHERS* (MW)	NET DEMAND MET (MW)	TOTAL GENERATION (MW)	NET TRANSNATIONAL EXCHANGE (MW) (+ve) Import, (-ve) Export
		(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I=A-(C+D))	(J=B+C+D+E+F+G+H)	(K)
0:00	50.04	144289	6213	15818	4632	6680	1730	114083	3108	123839	152264	-1042
0:15	49.99	143610	6236	15787	4631	6654	1736	114123	3114	123192	152281	-1069
0:30	49.94	143250	6241	16054	4631	6689	1761	112602	3084	122565	151062	-1079
0:45	49.94	142022	6262	16096	4631	6378	1744	111728	3073	121295	149912	-1114
1:00	49.94	141129	6225	16038	4631	6470	1735	110747	3045	120460	148891	-1034
1:15	49.94	139818	6234	15805	4630	6349	1743	109829	3102	119383	147692	-1074
1:30	49.89	139186	6235	15828	4630	6146	1743	110301	3108	118728	147991	-988
1:45	49.94	138981	6238	15633	4748	6081	1742	109405	3107	118600	146954	-1026
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